Statistics Problem Solutions (Problems 1–37)

Problem 1

Arithmetic Mean Formula: x■ = $\Sigma x_i / n$

Mean = 63139 / 20 = 3156.95 g

If first value = 500 g instead of 3265 g:

New Sum = 63139 – 3265 + 500 = 60374

New Mean = 60374 / 20 = 3018.7 g

Problem 2

Simple Mean = (42+51+50+40+60+54)/6 = 49.5

Weighted Mean = $\Sigma(w_i x_i) / \Sigma w_i$

= 11845 / 236 = 50.19

Problem 3

Median position = (n+1)/2

Median = (3245 + 3248)/2 = 3246.5 g

Problem 4

Mean = 97/9 = 10.78

Median = 8

If 35 becomes 65:

New Mean = 127/9 = 14.11

Median unchanged = 8

Problem 5

Mean ≈ 9.79 g/dl

Median ≈ 9.8 g/dl

Distribution approximately symmetric.

Problem 6

Mean ≈ 4.72 mm

Median = 4.1 mm

Mean > Median → positively skewed distribution.

Problem 7

Mode values:

Problem 4 → 8

Problem 5 → 9.4

Problem 6 → 2.8

Problem 8

Batsman A mean = 35

Batsman B mean = 35

Batsman A is more consistent because the scores vary less.

Problem 9

Method 1: Mean = 200, Median = 195, Range = 49

Method 2: Mean = 200, Median = 200, Range = 17

Method 2 shows lower variability.

Problem 10

Batsman A: Variance = 11.6, SD = 3.41

Batsman B: Variance = 522.8, SD = 22.87

Problem 11

Coefficient of Variation Formula: CV = (s / x■) × 100

CV■ = (12/80)×100 = 15%

CV■ = (0.20/0.50)×100 = 40%

Problem 12

Wages CV = 10%

Chocolate consumption CV = 25%

Chocolate consumption is more variable.

Problem 13

Arithmetic Mean = 227/12 = 18.9

Log-transformed mean ≈ 11.1

Problem 14

After normalization of judge scores,

Winning contestant = Contestant 6.

Problem 15

$Z = (85 – 70) / 12 = 1.25$

$P(Z > 1.25) \approx 0.106$

$\approx$ 10.6% of students scored above 85.

Problem 16

Reference Range = Mean $\pm$ 1.96 SD

$\approx$ 6.26 – 13.32 g/dl

Problem 17

Reference Range = $2.74 \pm 1.96(2.60)$

$\approx$ 0 – 7.84 mmol/L

Problem 18

Standard Deviation Formula: $s = SE\sqrt{n}$

Group 1 SD $\approx$ 0.40

Group 2 SD $\approx$ 0.75

Group 3 SD $\approx$ 1.89

Problem 19

Standard Error = $1.8 / \sqrt{48} \approx 0.26$

95% CI = 9.79 ± 0.51

CI = 9.28 – 10.30

Problem 20

99% CI = 9.79 ± 0.67

CI = 9.12 – 10.46

Problem 21

95% CI = 9.28 – 10.30

For $n < 30$ use t■distribution.

Problem 22

SE = $5/\sqrt{3} = 2.89$

Z = 0.69

Probability ≈ 0.754

Problem 23

Possible outcomes: HH, HT, TH, TT

P(0 heads) = 1/4

P(1 head) = 1/2

P(2 heads) = 1/4

Problem 24

H■: $\mu = 120$

H■: $\mu < 120$

Problem 25

Type I Error: Concluding cholesterol is higher when it is not.

Type II Error: Failing to detect higher cholesterol.

Problem 26

t = –2.08

Reject H■ at 0.05 level but not at 0.01.

Problem 27

t = 1.58

Since 1.58 < 1.833 → Fail to reject H■.

Problem 28

t = –2.12

Reject H■.

Problem 29

Z = –3.0

Reject H■.

Problem 30

$t = -2.54$

Drug significantly reduces infarct size.

Problem 31

$t = -1.32$

Result is borderline significant.

Problem 32

95% CI = 179.45 – 190.55

Since 190 lies within the interval, result not significant.

Problem 33

Sample size $n \approx 142$

Problem 34

Sample size $n \approx 36$

Problem 35

Required sample size ≈ 32

Problem 36

Paired $t \approx 3.56$

ICU treatment significantly increases oxygen saturation.

Problem 37

$Z \approx 5.79$

Difference in arterial pressure highly significant.